EchoScribe: An AI-Powered Clinical Documentation Platform with Multi-Provider Speech Processing, Speaker Diarization, and Intelligent SOAP/GROW Note Generation

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***Abstract*—Therapists and academic mentors spend much of their working time writing session records. Language models can draft these records from a recording, but a draft may state things the session never contained, and a single cloud model is a single point of failure for quota, outage and data-governance reasons. We present EchoScribe, an open system that turns a recorded two-person session into a SOAP (therapy) or GROW (mentoring) note in which every sentence cites the transcript turns that support it. A model other than the writer, from another vendor when one is configured, checks each sentence against its cited turns, and notes that fail are regenerated on another model. An embedded multi-provider router keeps every processing role available through quota exhaustion and outages, and speaker separation runs locally on the original recording, with each word assigned to a speaker by its timestamp. On the 40 test dialogues of ACI-Bench, citing evidence halved unsupported content against a single-pass note from the same writer (11.2% to 5.0% of sentences) without lowering overlap with reference notes, and the verifier caught 98% of planted errors. On 57 mock consultations, speaker-attributed word error was 23.8% with word timestamps and 53.0% without. Clinician or mentor identification needed no enrolment and was correct in 57 of 57 consultations and 40 of 40 counselling dialogues. In fault-injection simulations, a last-known-good routing strategy cut 95th-percentile latency from 31.7 s to 9.1 s.**

***Index Terms*—clinical documentation, hallucination detection, speaker diarization, large language models, LLM routing, SOAP notes, GROW coaching**

I. Introduction

Writing session notes is one of the most time-consuming non-clinical tasks of mental health work and of academic mentoring. A SOAP note records the client's account, the clinician's observations, an assessment and a plan [1]; a GROW note records a mentee's goal, current reality, options and way forward [2]. Documentation extends the working day [3], and ambient scribes that draft notes from recorded encounters are now in clinical use and under randomized evaluation [4], [5].

Two gaps limit what such systems can be trusted with. A note may contain statements the conversation never supported [6], [7], and the reviewer cannot tell which sentence rests on what. And published scribes depend on one model from one vendor, so a quota, an outage or a retired model stops documentation altogether; neither gap is addressed for counselling or mentoring [8]–[10]. EchoScribe's contributions are:

1) **Evidence-linked notes with independent verification.** The writer cites turn numbers for every sentence; a model other than the writer, preferably from another vendor, labels each sentence supported, partially supported or unsupported; notes that fail a threshold are regenerated on another model, and flagged sentences can be repaired from the transcript (Section III-C).
2) **An embedded multi-provider router** serving every processing role with routing strategies, per-key exponential cooldown and a provider-wide breaker, adapted from the open-source OmniRoute gateway [11] (Section III-A).
3) **Local speaker attribution without enrolment:** pyannote diarization of the original recording, words assigned

to turns by their timestamps, and role resolution from eight conversation statistics and a model check (Section III-B).

4) **A reproducible evaluation** on public data (ACI-Bench, AnnoMI, PriMock57, LibriSpeech simulations): router under faults, grounding against a single-pass baseline, verifier accuracy, role identification, and speech under noise (Section IV).

II. Related Work

Ambient scribes and note generation. Note generation from doctor–patient dialogue has been benchmarked on ACI-Bench [10] and in MEDIQA-Chat [12]; PriMock57 adds audio of mock consultations [13] and AnnoMI counselling dialogues [14]. Adapted models can match experts on clinical summarisation [15], [16], and deployed systems report physician review [8], [9]. For mental health, Adhikary et al. benchmarked summarisation of counselling sessions [17]. None of these produces mentoring (GROW) notes, links each sentence to its evidence, or survives its vendor failing.

Hallucination and verification. Hallucination is a recurring failure of generated text [6], and clinical safety frameworks annotate hallucinations and omissions [7]. Remedies include citation-constrained generation [18], atomic-fact scoring [19], self-consistency sampling [20] and self-verification [21]. EchoScribe combines citation-constrained writing with sentence-level verification by an *independent* model, so the checker does not share the writer's failure modes, and uses the verdicts for review and regeneration.

Speech. Diarization combines activity detection, speaker embeddings and clustering [22]: ECAPA-TDNN [23] with spectral clustering [24] is a strong baseline, while pyannote's powerset model [25] and Sortformer [26] learn segmentation directly; WhisperX aligns recognition with diarization [27], and speaker roles in clinical conversations have been studied [28]. SepFormer [29] in SpeechBrain [30], trained on WHAM! [31], is one of many enhancement models [32], but enhancement artefacts can hurt recognition [33].

Routing. RouteLLM routes between a strong and a weak model [34], FrugalGPT cascades models to cut cost [35], and gateways such as OmniRoute [11] provide combo routing. EchoScribe embeds this per role and escalates a note when its content fails, not only when a call does.

III. System Design

Fig. 1 shows the pipeline. A browser client records or uploads audio, an Express.js server coordinates every model call, and Supabase stores records under per-user row-level security. Five roles call language or speech models: STT, DIARIZE, ROLE_LABEL, NOTE_SYNTH and VERIFY.

A. Embedded multi-provider router

Each role is served by a *combo*: an ordered list of targets $\langle$vendor, model, keys$\rangle$ and a routing strategy. Vendors are thin adapters: Gemini [36], Groq-hosted gpt-oss-120b [37] and Whisper [38], Deepgram and a local pyannote service ship with the system. The strategy orders targets per request: *priority* (declared order; the next target is used only when every key of the current one is cooling down or retired, or the model itself fails), *round-robin*, *random*, *strict-random*, *least-used*, and *lkgp* (the last target that succeeded goes first).

Failures are classified. A key that fails authentication is retired; a rate-limited or failing key enters an exponential cooldown of $5 \cdot 2^{k-1}$ s (capped at 5 min) after its k-th consecutive failure, reset on success; a vendor with 15 failures within 30 min is skipped for 10 min (2 in 5 min for the local service); an HTTP 400 is treated as specific to the request and moves on without cooling the key. These mechanisms and constants are adapted from OmniRoute [11]. When every target fails the API returns HTTP 503 with the attempts. Keys stay in the user's browser unless the user syncs one, which is then stored encrypted.

B. Speech processing and role resolution

The recording is not denoised: SepFormer enhancement, our first design, raised word error rates (Section IV), as reported for other front ends [33]. DIARIZE runs pyannote community-1 [25], [39] locally on the original 16 kHz recording, on a GPU when present, with two speakers and non-overlapping turns; it falls back to cloud audio diarization, then to splitting the transcript with a language model. STT transcribes the original recording; when the recogniser returns word timestamps (Whisper, Deepgram), each word goes to the turn that contains it [27], otherwise text is spread over turns by duration.

Roles are resolved without voice enrolment. For each speaker the system computes eight statistics: speaking time, word count and mean turn length, questions, professional vocabulary, guiding phrases, personal and emotional phrases, number of turns, and who spoke first. ROLE_LABEL receives the transcript with these statistics and must assign exactly one professional and one client; invalid answers move to the next target, and a deterministic default is used only if every target fails.

C. Evidence-linked notes, verification and repair

The labelled transcript is numbered into turns $T_1 \dots T_n$. NOTE_SYNTH must end every sentence of the note sections, and of the risk notes, with the turns that support it ("…poor sleep for two weeks [T4, T6]"), write one fact per sentence, and write "Not discussed" rather than an uncited sentence. VERIFY receives each sentence with only its cited turns and labels it *supported*, *partial* (adds a detail the turns lack) or *unsupported*. Verifiers are ordered another vendor first, then another model of the writer's vendor, and never the writer's own model. If the unsupported fraction exceeds $\tau = 0.2$, the note is regenerated with the writer's model moved to the end of the combo, re-verified, and the draft with fewer unsupported sentences is kept.

Two further steps work at sentence level. *Repair*: each flagged sentence goes back to a writer model with the transcript and the verifier's reason to be rewritten with citations, or

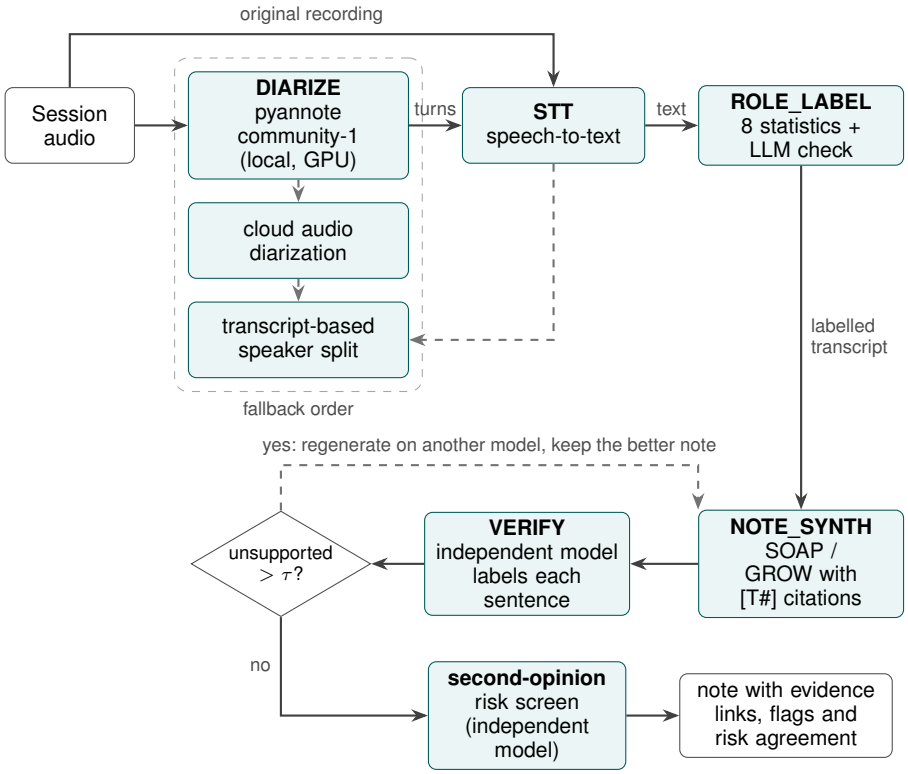


Fig. 1: EchoScribe pipeline. Shaded boxes are processing roles, each served by a combo of the embedded router (Section III-A). Speaker separation falls back from local pyannote to cloud audio diarization to a transcript-based split (dashed). The note writer cites transcript turns; an independent model (another vendor, else another model) verifies each sentence, and notes over the unsupported threshold τ are regenerated on another model. A second model screens independently for risk.

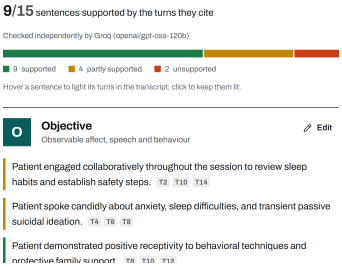


Fig. 2: EchoScribe's session summary for a scripted example session (invented transcript; real pipeline output). The verification bar counts sentences the independent verifier (here gpt-oss-120b, checking a Gemini-written note) judged supported, partly supported or unsupported; Objective sentences carry a stripe for their label (green supported, amber partly, red unsupported) and the turns they cite.

removed if the transcript does not support it; rewrites are verified again. *Two-judge agreement*: a flag stands only if a second judge, chosen to differ from the first and from the writer, also flags it. These steps are evaluated in Section IV.The interface shows each sentence's evidence on hover and marks partial and unsupported sentences (Fig. 2); an *Audio* button downloads the saved recording.

D. Risk screen with a second opinion

In therapy mode the writer also screens for suicidal ideation and self-harm; in mentoring mode, for burnout and severe distress. A second model, chosen like the verifier, repeats the screen independently and cites its evidence turns. The flag is raised if either model raises it, favouring sensitivity, and disagreement is shown as "models disagree — review the transcript". The screen is a model judgement, not a validated instrument such as the C-SSRS or PHQ-9 [40], [41], and the interface states that the clinician remains responsible for risk assessment.

IV. Evaluation

Every number below is generated from result files by scripts in the public repository (`eval/`); only free-tier model access was used.

A. Setup

Data. Notes are evaluated on the 40 test dialogues of ACI-Bench (test1, CC BY 4.0) [10], whose doctor–patient visits have reference notes, and, without reference notes, on 40 high-quality motivational-interviewing dialogues of AnnoMI [14] drawn with a fixed seed over 32 topics. Real conversational speech comes from PriMock57 [13]: 57 mock primary-care consultations (8.64 h) acted by clinicians and employees, with manual transcripts (CC BY 4.0). Controlled noise tests use 12 simulated two-person conversations built from LibriSpeech test-clean [42] (ten alternating utterances, 0.3–0.9 s gaps, exact reference turns), clean and with babble plus pink noise at 5 dB

TABLE I: Router in the mixed-fault scenario (mean of 5 seeds): success rate, failed provider calls per request, 95th-percentile latency.

Policy	Success	Failed calls/req.	p95 (s)
No memory	99.93%	0.497	31.8
Fixed 60 s	98.14%	0.147	31.5
Priority	99.86%	0.258	31.7
Round-robin	100.00%	0.092	31.4
LKGP	100.00%	0.097	9.1

and 0 dB SNR. **Models and metrics.** Gemini models [36] are used in rotation across three keys; gpt-oss-120b [37] and Whisper large-v3 [38] are served by Groq. We report ROUGE [43] and BERTScore [44] against reference notes, WER after text normalisation, DER with a 0.25 s collar [45], and cpWER, the speaker-attributed WER of CHiME-6 [46]. Intervals are Wilson 95%; differences are tested with a paired bootstrap (10,000 resamples).

B. Router under injected faults

We drive the router's own code with simulated vendors on a virtual clock: one hour of Poisson traffic at 10 requests/min over three vendors, in three scenarios: a per-key quota below the offered load, a 15-minute outage of the preferred vendor, and a *mixed* case (revoked key, quota, a fallback timing out 20% of the time). We compare walking the chain with no memory, the fixed 60 s suspension of our earlier design, and three strategies (5 seeds, Table I).

Where a healthy fallback exists no policy changes success much, so differences lie in wasted calls and latency. Last-known-good stops returning to a vendor that keeps timing out and cuts mixed-scenario p95 latency from 31.7 s to 9.1 s (difference $[-28.3, -11.7]$ s, $p < 0.001$, faster in all 5 seeds). Under a per-key quota, round-robin and LKGP wasted 0.014 and 0.033 failed calls per request against 0.227 for priority; exponential cooldown alone is not uniformly better, because its 5 s first step re-probes an exhausted key. Priority keeps the preferred model at the cost of probing; LKGP minimises latency but can stay on a fallback after the preferred model recovers.

C. Evidence-linked notes and independent verification

Does citing evidence reduce unsupported content? With the same Gemini writer and prompt apart from the citation rule, we wrote a single-pass SOAP note and an evidence-linked note for each of the 40 dialogues, shuffled the sentences of both, and had a judge from another vendor (gpt-oss-120b) label each against the *full* transcript without knowing its source (Table II). Evidence-linked notes had fewer unsupported sentences (5.0% against 11.2%, non-overlapping intervals), fewer notes with any unsupported sentence (20/40 against 30/40), and no lower ROUGE or BERTScore.

Two writers and the cascade. Judged against only their cited turns, the stricter test the interface uses, Gemini- and gpt-oss-written notes had citation coverage 95.6% and 97.3%, with 83.9% and 77.0% of sentences supported (15 of the Gemini notes were checked cross-vendor, the rest cross-model, after Groq's daily cap). With $\tau = 0.2$, 1 of 40 Gemini-first notes would have been regenerated.

TABLE II: Notes on ACI-Bench test1. Top: single-pass vs evidence-linked notes from the same writer, every sentence judged blind against the full transcript by another vendor. Bottom: planted errors caught by the verifier, and false alarms on unchanged sentences.

	Single-pass	Evidence-linked
Supported (95% CI)	80.4% (77.5–83.0)	88.4% (85.9–90.6)
Partly supported	8.3%	6.6%
Unsupported (95% CI)	11.2% (9.2–13.6)	5.0% (3.6–6.8)
Notes with an unsupported sentence	30/40	20/40
ROUGE-1 / 2 / L	44.3 / 16.2 / 41.5	45.4 / 16.9 / 42.5
BERTScore F1	47.1	48.0
	Gemini-written	gpt-oss-written
Changed number caught	95%	95%
Flipped negation caught	100%	100%
Borrowed sentence caught	100%	100%
False alarms on unchanged	7%	12%

Does the verifier catch errors? We planted three kinds of error into sentences the verifier had accepted (a changed number, a flipped negation, and a sentence borrowed from another dialogue's note with its citations) and kept other accepted sentences as controls. Across 109 planted errors the verifier flagged 98% as partial or unsupported, with 9% false alarms on 129 unchanged sentences (Table II, bottom); controls had already been accepted, so this measures consistency, not precision on fresh sentences.

Counselling dialogues. On AnnoMI, evidence-linked notes (without sentence repair) for 40 dialogues had 549 sentences, 96.4% cited, of which 80.3% were judged supported, 13.3% partly and 6.4% unsupported. Judged again by the held-out Groq model, 65.4% of the 549 sentences were supported and 10.6% unsupported; the first figure was inflated because a Gemini model judged notes that a Gemini model had written.

One fact per sentence and repair. The writer is asked for one fact per sentence; each sentence the judge flags goes back to a writer model with the transcript and the reason, to be rewritten from the evidence or removed, and rewrites are verified again (44 of 60 flagged sentences were rewritten, 12 removed). Because the pipeline's own judge also chose the sentences to repair, we judged the notes with a held-out judge, Groq gpt-oss-120b (a different vendor from the Gemini writer), against the turns each sentence cites, for the earlier notes and the final notes of the same 40 dialogues. Earlier notes had 79.1% of 727 sentences supported (76.0–81.9) and 7.1% unsupported, with 25 notes holding an unsupported sentence; final notes had 88.5% of 893 supported (86.2–90.4) and 3.2% unsupported (2.3–4.6), with 16 notes affected. The mean per-note gain in supported sentences was 9.45 points (paired bootstrap 95% interval 6.2–12.8; 30 notes improved, 9 got worse). The pipeline's own judge was more generous, rating 93.4% supported as written and 98.5% after repair, and agreed with the held-out judge on 88.5% of the final sentences; we therefore report the held-out figures. Overlap with the reference notes also rose (ROUGE-1/2/L 53.0/21.8/49.4, BERTScore 49.8, against 45.4/16.9/42.5 and 48.0 before these

TABLE III: Speech results (%). Top: simulated conversations (12 × 3 conditions): recognition with and without SepFormer denoising, and DER. Middle: speaker-attributed cpWER of each path; the first row is our first design, the bold row the current one. Bottom: PriMock57 mock consultations.

	Clean	5 dB	0 dB
WER Whisper, original / denoised	2.1 / 2.9	13.5 / 28.9	36.0 / 70.9
WER Gemini, original / denoised	5.9 / 4.0	16.9 / 26.9	45.5 / 65.0
DER pyannote / SpeechBrain	9.3 / 23.1	12.9 / 25.1	22.6 / 24.9
SepFormer + SpeechBrain + Gemini	43.4	48.8	77.9
pyannote + Whisper, by duration	21.4	47.1	68.8
pyannote + Whisper, by word time	**2.6**	**19.7**	**53.2**
SpeechBrain + Whisper, by word time	21.0	27.1	46.9
Cloud audio diarization (Gemini)	17.8	37.1	59.3
PriMock57, 57 consultations, 8.64 h (pyannote + Whisper)			
DER / WER	16.8% / 18.8%		
cpWER by word time / by duration	23.8% / 53.0%		

steps).

D. Role identification

With speaker labels replaced by random identifiers, the eight statistics alone named the clinician correctly in 38 of 40 ACI-Bench dialogues, the model alone in 37, and their combination, as deployed, in 39 (97.5%). Only 2 ACI dialogues had the patient speaking more than the doctor. In AnnoMI counselling the client often talks more: statistics alone were right in 38 of 40 dialogues, the model alone in 40, and the combination in 40; in the 17 dialogues where the client spoke more, the three methods were right in 16, 17 and 17. On PriMock57 the deployed combination was right in 57 of 57 consultations.

E. Speech recognition and speaker attribution

Table III reports the speech results. Denoising before recognition, as in our first design, raised word error rates under noise for both recognisers (Whisper at 0 dB from 36.0% to 70.9%; $p < 0.001$ and $p = 0.006$ for Whisper and Gemini), so the deployed pipeline transcribes the original recording. pyannote separated speakers with 9.3% DER on clean audio against 23.1% for SpeechBrain clustering ($p < 0.001$; at 5 dB $p < 0.001$). Speaker attribution depends as much on placing words as on finding turns: spreading the transcript over pyannote's turns by duration gave 21.4% cpWER on clean audio, and assigning each word to the turn containing its Whisper timestamp gave 2.6% ($p < 0.001$ in all three conditions). Against cloud audio diarization the local pipeline attributed better on clean audio ($p = 0.032$) and at 5 dB (19.7% against 37.1%, $p = 0.005$). At 0 dB pyannote's DER rises to 22.6% ($p = 0.345$ against SpeechBrain), and SpeechBrain clustering with word timestamps attributes slightly better (46.9% against 53.2%); the difference from cloud diarization at 0 dB is not significant ($p = 0.520$). On real mock consultations the same pipeline reached 16.8% DER and 18.8% WER, with 23.8% speaker-attributed error against 53.0% when words were spread by duration.

Per minute of audio, pyannote took 2.1 s on a laptop GPU (RTX 4050) and 22.0 s on its CPU, against 78.2 s for the first design's SepFormer enhancement with SpeechBrain diarization; Whisper on Groq took 10.2 s and Gemini speech-to-text 67.3 s.

F. GROW notes on mentoring sessions

Volunteers role-played mentoring sessions from a script we provided: 19 recordings (277 min; 3 mentee personas) from the volunteers and 4 (32 min) made by the authors; none is clinical or real mentee data. We ran all of them through the deployed pipeline end to end. pyannote found two speakers in 19/19 and 4/4 sessions, role resolution assigned one mentor and one mentee in 19/19 and 4/4, and 19/19 and 4/4 notes filled all four GROW sections (volunteer and author sessions, respectively). The pipeline's judge rated 81.6% of 283 and 94.2% of 52 sentences supported; the held-out Groq judge rated 82.7% and 88.5%. Sessions share personas, no human rated GROW content, and the speech is acted, so these figures show feasibility, not accuracy.

V. Discussion and Limitations

Deployment. EchoScribe runs on the clinician's machine; only the configured model vendors receive audio or text, so an institution can restrict processing to vendors its governance policy approves. Sentence-level flags give a reviewer a concrete place to look instead of an unexplained confidence score [47].

Limitations. (i) ACI-Bench and PriMock57 are medical, and AnnoMI has no reference notes; no public corpus of counselling or mentoring sessions with reference notes exists, and the mentoring sessions we ran are acted and share personas. (ii) Support labels come from a model; the planted-error test bounds how often it misses errors of three kinds, but subtler omissions and misattributions need human review [7]. (iii) Simulated and mock speech has no overlap or strong emotion, and at 0 dB our attribution is not better than cloud diarization. (iv) Results depend on model versions that vendors retire. (v) The risk screen has not been validated against clinician labels. Blinded rating kits for notes and verifier agreement are released with the code. **Ethics:** no patient or clinical data were used; experiments use public datasets, simulated speech and scripted role-plays recorded by volunteers who knew the audio would be processed by the system, and free-tier model vendors may use submitted text for training, so deployments with real sessions should use institutionally approved vendors or local models.

VI. Conclusion

EchoScribe produces SOAP and GROW notes from recorded two-person sessions in which every sentence points to the turns that support it and is checked by a model other than its writer, with a two-model risk screen and an embedded router that keeps every role available through quota exhaustion and outages. Citing evidence halved unsupported content against a single-pass note from the same writer; assigning words to speakers by timestamp cut speaker-attributed error from 53.0% to 23.8% on mock consultations; and role identification needed no enrolment. The code, evaluation

scripts and rating kits are available at https://github.com/Aravind-Krishnan-S/ECHOSCRIBE.

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